jal-netra_ORDERING

Jal-Netra (engineering marvel): ordering guide (what to buy, where, in what order)

> Companion to MANUFACTURING_MANUAL.md. Main caveat for this project: **Gate 1 acoustic test first; not sellable**. Prices and stock are from LCSC on the run date; vendors, shipping and customs rules change: verify before paying. General rules and vendor list: manuals/ORDERING_GUIDE.md.

1. Order of work

1. Read the manual and the caveat above.
2. Order the circuit boards and SMD assembly (section 2).
3. While the boards travel, order the other items (sections 3 and 5) and make the printed or laser-cut parts.
4. Assemble, program and test only after everything has arrived; follow the manual.

2. Order the circuit boards (JLCPCB, PCB + assembly)

Files (all in this repository):

- Gerbers: `utility10/pcb/release/jalnetra/1_fabrication/jal_netra_gerbers.zip`
- Assembly BOM: `utility10/pcb/release/jalnetra/2_assembly/jal_netra_JLC_BOM.csv`
- Pick-and-place (CPL): `utility10/pcb/release/jalnetra/2_assembly/jal_netra_JLC_CPL.csv`
- Full BOM with LCSC numbers and prices: `costing/jalnetra_priced_BOM.csv`

Steps:

1. Create an account at **jlcpcb.com** and click *Order now*. Upload the Gerber ZIP. The viewer must show the board outline, both copper layers, silkscreen and drill holes.
2. Settings: **2 layers, 1.6 mm, FR-4**, lead-free HASL (use **ENIG** if the board has an LGA/QFN fine-pitch part, e.g. ADXL345, ESP32 module pad, BQ25570), green mask, white silk, quantity 5 (the minimum), *Remove order number: specify a location* if you want to avoid the order-number mark. Heavier copper (2 oz) only where the board notes say so.
3. Switch on **PCB Assembly**: *Economic*, assembly side *Top*, and set tooling holes *Added by JLCPCB* unless the board already has them. Choose the number of boards to assemble (2 or 5).
4. Upload the BOM and CPL. In the parts step, every line must match (LCSC numbers are in the BOM). Where a part is shown as *out of stock* or *no match*, use the alternative suggested by JLCPCB only after checking footprint and value against the list below; do not accept a different footprint.
5. In the **component placement preview**, check pin 1 / polarity of every IC, diode, LED, electrolytic capacitor, USB-C and connector against `*_assembly_top.pdf` in the same folder. Wrong rotation is the most common assembly error: ask JLCPCB support to rotate before approving.
6. Pay, then approve the assembly engineering review messages (they usually ask about a part or rotation). Delivery by DHL/FedEx/UPS takes about 1 to 2 weeks to India.

Parts **not** assembled by JLCPCB (you buy and solder them yourself):

Refs	Value	Part	LCSC
J6	Transducer A (BTL)	JST B2B-PH-K-S(LF)(SN)	C131337
J7	Transducer B (BTL)	JST B2B-PH-K-S(LF)(SN)	C131337
J4	OLED I2C	JST B4B-PH-K-S(LF)(SN)	C131334
J5	UART host / BLE	JST B4B-PH-K-S(LF)(SN)	C131334
J3	SWD	HCTL PZ254-2-03-Z-8.5 (ZHOURI C5116479 had only 340 in stock)	C2894964
TP1, TP2, TP3, TP4, TP5, TP6	TP		
BZ1	Buzzer	YUEXIN KC1206 16 ohm passive magnetic 12x9.5 mm (2048 Hz)	C781869

Order these with the same LCSC order (add them to the cart so one shipment arrives) or buy locally in India (section 5).

Parts cost and stock (live LCSC prices on the run date)

Total parts per board: **US10.10*for1board* * *, * * *US*9.01 each for 10 boards**, **US$7.66 each for 100 boards** (about INR 889 / 793 / 674 at 88 INR/USD). The per-line break-up is in `costing/jalnetra_priced_BOM.csv`. This is parts only: PCB fabrication, assembly fees, shipping, customs and GST come on top (see `manuals/ORDERING_GUIDE.md`, section 6).

Stock warnings on the run date:

- TP (no lcsc price (hand-fit or not priced), stock)

Refs	Qty	Value	Part	LCSC	Unit USD (1 board)	Line USD
U1	1	STM32G431CBT6	STM32G431CBT6 (LQFP-48)	C529355	4.8168	4.817
U4,U7	2	TS5A3159	TI TS5A3159DCKR	C46388	0.3784	0.757
U5,U6	2	OPA356	TI OPA356AIDBVR	C183100	1.3886	2.777
U3	1	AP2112K-3.3	Diodes AP2112K-3.3TRG1 (from the LCSC datasheet link)	C51118	0.1814	0.181
FB1	1	BLM18 600R	Murata BLM18AG601SN1D	C19330	0.0122	0.012
Y1	1	8 MHz	YXC X32258MOB4SI 8 MHz 12 pF 3225 (matches the 15 pF load caps)	C2682775	0.1994	0.199
D5,D6	2	BAV99	JSCJ BAV99	C68983	0.0168	0.034
J6	1	Transducer A (BTL)	JST B2B-PH-K-S(LF)(SN)	C131337	0.0359	0.036
J7	1	Transducer B (BTL)	JST B2B-PH-K-S(LF)(SN)	C131337	0.0359	0.036
R30,R31,R32,R33	4	22	UNI-ROYAL 0603WAF220JT5E	C23345	0.0017	0.007
C20,C21,C28,C30	4	10n	Samsung CL10B103KB8NNNC 0603 10nF X7R 50V	C1589	0.0075	0.030
R35,R36,R17,R7,R5	5	1k	0603 1k 1%	C21190	0.0017	0.008

Refs	Qty	Value	Part	LCSC	Unit USD (1 board)	Line USD
C22,C23,C24,C1,C2,C3,C4,C33,C37	9	100n	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	0.0075	0.068
C25,C27,C29,C31,C38,C39	6	1u	0603 1uF X7R	C15849	0.0144	0.086
R10,R11,R21,R22	4	10k	0603 10k 1%	C25804	0.0024	0.010
R12,R16	2	100k	0603 100k 1%	C25803	0.0030	0.006
R13	1	2.2k	0603 2.2k 1%	C4190	0.0018	0.002
R14	1	240	UNI-ROYAL 0603WAF2400T5E (JLCPCB listing)	C23350	0.0024	0.002
R15,R19	2	0	UNI-ROYAL 0603WAF0000T5E	C21189	0.0022	0.004
R18	1	100	UNI-ROYAL 0603WAF1000T5E	C22775	0.0024	0.002
R20	1	33	UNI-ROYAL 0603WAF330JT5E	C23140	0.0024	0.002
C32	1	22p	Samsung CL10C220JB8NNNC C0G	C1653	0.0090	0.009
C34	1	4.7u	Samsung CL10A475KP8NNNC X5R 10V	C1705	0.0128	0.013
C35,C36	2	15p	Samsung CL10C150JB8NNNC C0G	C1644	0.0112	0.022
J4	1	OLED I2C	JST B4B-PH-K-S(LF)(SN)	C131334	0.0458	0.046
R23,R24	2	4.7k	0603 4.7k 1%	C23162	0.0028	0.006
J5	1	UART host / BLE	JST B4B-PH-K-S(LF)(SN)	C131334	0.0458	0.046
J3	1	SWD	HCTL PZ254-2-03-Z-8.5 (ZHOURI C5116479 had only 340 in stock)	C2894964	0.0766	0.077
SW2	1	Test / zero	Omron B3U-1000P	C231329	0.1894	0.189
TP1,TP2,TP3,TP4,TP5,TP6	6	TP				
J1	1	USB-C 5V	TYPE-C-31-M-12	C165948	0.1890	0.189
F1	1	PTC 500mA	Littelfuse 1206L050YR (6 V)	C163512	0.0742	0.074
D3	1	SMAJ5.0A	Littelfuse SMAJ5.0A	C83329	0.1153	0.115
R1,R2	2	5.1k	0603 5.1k 1%	C23186	0.0016	0.003
C5	1	10u	0805 10uF X5R	C15850	0.0441	0.044
BZ1	1	Buzzer	YUEXIN KC1206 16 ohm passive magnetic 12x9.5 mm (2048 Hz)	C781869	0.1513	0.151
Q1	1	MMBT3904	MMBT3904	C20526	0.0118	0.012
D2	1	1N4148WS	1N4148WS	C2128	0.0120	0.012
D1	1	Green	Hubei KENTO KT-0603YG yellow-green 570 nm, Vf about 2 V (lights well from 3.3 V with 1 k; the Everlight 19-217 green was out of stock)	C2289	0.0105	0.011

2b. Enclosure: printed parts and fasteners (generated from the board)

- Print `4_enclosure/` base and lid (outer about 101 x 81 x 22 mm) in PETG or ABS, 3 perimeters, 25 % infill; the lid closes with **4 x M2.5 x 10 mm self-tapping screws** into the corner columns; the board sits on its standoffs with 4 x M2.5 x 6 mm self-tapping screws.
- Print `*_enclosure_caps.stl` (light pipe, plunger): plunger for SW2; light pipe for D1. Light pipes in clear PETG or resin; fit plungers and pipes from inside before closing the lid.
- indoor, dry use; print in PETG or ABS, 3 perimeters, 25 % infill
- Print one set first and test-fit the board, the holder and every cap; the dimensions are first-pass.

3. Bought-in and mechanical items

See the parts and materials section of the manual.

4. Confidence (read before you order)

- **Order confidence: Medium-High**: limited stock on: BZ1 Buzzer (stock 140).
- **Works-as-intended confidence: Low-Medium**: Clamp-on ultrasonic transit-time measurement has a hard analog front end; the manual itself says Gate 1 acoustic test comes first and it is not sellable.
- **First thing to do:** Run the Gate 1 acoustic test with the evaluation setup before any more boards.
- Rubric and the reasons for every project: `costing/CONFIDENCE.md`. These are judgements, not measurements: nothing here has been built.

5. Where to buy and whom to approach

Item type	India (fast, GST invoice)	Overseas (cheaper, 1 to 3 weeks, customs)
Printed circuit board and SMD assembly	JLCPCB is the practical choice for 5 to 100 boards; local PCB shops quote per project (ask for the Gerber ZIP price)	**JLCPCB** (jlcpcb.com), PCBWay (pcbway.com)
Connectors, headers, passives, ICs by LCSC number	element14 India, Mouser India, DigiKey India (genuine, GST invoice, higher price); local electronics markets for common parts	**LCSC** (lcsc.com) with the same cart as JLCPCB
Modules: ESP32 dev boards, SIM7080G, LoRa (RA-08 / RAK3172), pH / EC / DO modules, relay modules, load cells, ultrasonic sensors	Robu.in, Robocraze, ThinkRobotics, Sunrom Electronics, Amazon.in	AliExpress, Seeed Studio, Waveshare
Motors, stepper drivers, wheels	Robu.in, Robocraze, local robotics shops	StepperOnline (omc-stepperonline.com)
Li-ion / LiFePO4 cells and packs, solar panels, MPPT controllers	Authorised battery and solar dealers, Amazon.in (buy protected cells with an invoice)	not recommended by post (shipping rules for batteries)
Laser-cut and 3D-printed parts	Local laser-cutting and 3D-printing shops (search IndiaMART, Justdial or maker spaces): send the SVG/DXF/STL from the repository and ask for a quote per piece	JLC3DP (jlc3dp.com), PCBWay 3D printing
Fasteners, tubing, hardware, enclosures	Local hardware, Robu.in, Amazon.in	n/a

How to approach a vendor: send the project name, the exact file (BOM, CPL, Gerber ZIP, STL or SVG), the quantity, and ask for the price, the lead time, the GST invoice and the warranty or return terms. Keep the reply for your records.